



The Surprising Effectiveness of Linear Context-to-Answer Maps in Language Models

A context-conditioned metamodel of answer activations

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TL;DR

- A single **linear map** predicts an LLM’s mean answer activation from its final context activation **before generation**: held-out R^2 **0.75**, and rank-1 retrieval of the true answer among 10,000 candidates at **0.98**
- It is already **~75% present in the base model**, then improves significantly with SFT, and barely changes with DPO and RLVR
- It is **shared between the assistant in chat template, assistant in stories, and other fictional characters in stories**, providing evidence for the persona selection model
- It can be used to predict **sycophancy, hallucination and misalignment before inference**.

MOTIVATION

- LLMs can be viewed as deterministic maps from *contexts* to *answer distributions*
- A **simple approximation** of that map would be extremely useful: much easier to analyze theoretically, and
- But it need not exist: the map is billions of parameters applying dozens of nonlinear layers.
- Prior work either uses simple but *narrow* predictors (specific tokens, activations, behaviors), or distillation, a complex whole-distribution predictor
- **Our question**: can a **simple** approximation capture the mapping *as a whole*?

THE RESIDUAL CONTEXT-ANSWER MAP

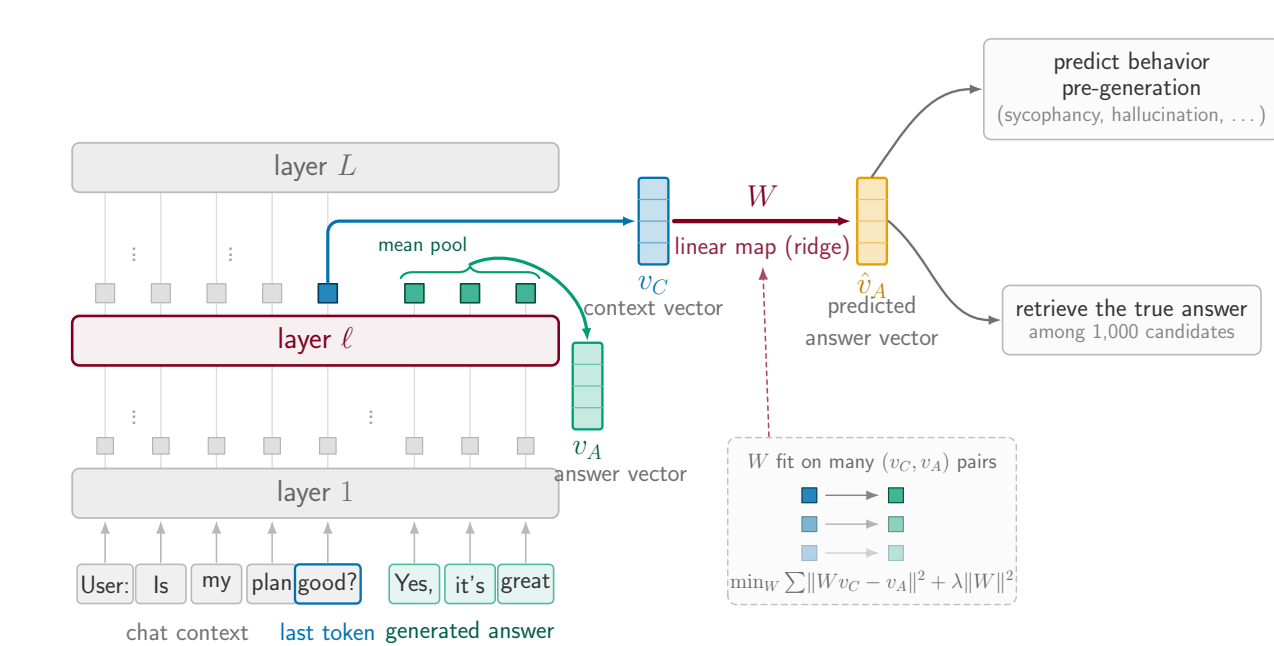


Figure 1. v_C = residual state at the last context token (layer ℓ); v_A = mean answer-token state. Fitted maps $v_C \mapsto v_A$; readouts consume $\hat{v}_A$.

1. THE MAPPING HAS HIGH PREDICTIVE POWER AND IS MOSTLY LINEAR

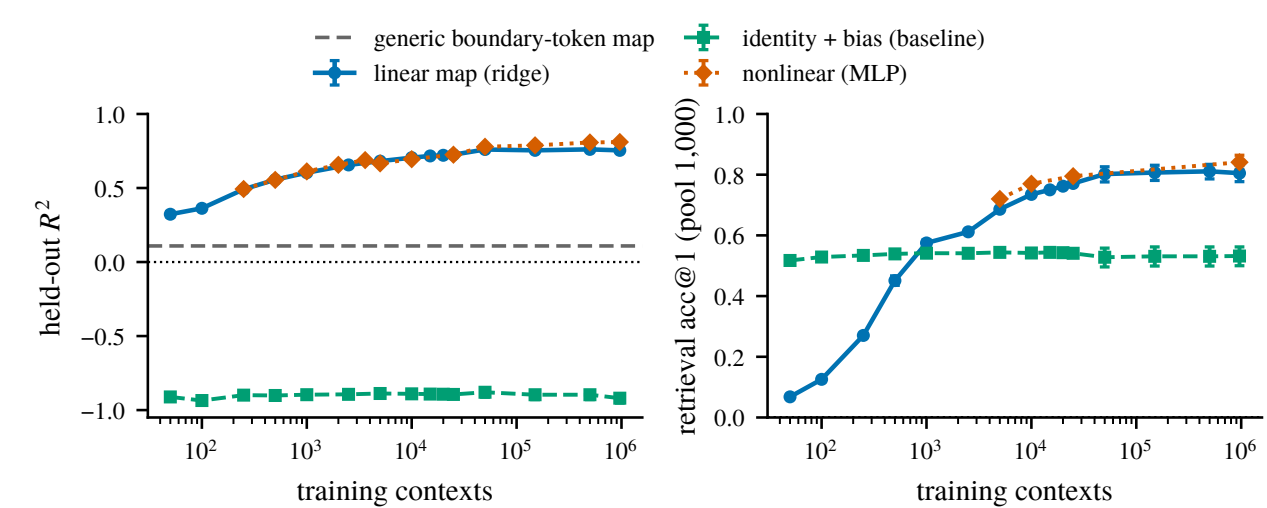


Figure 2. Held-out R^2 (left) and rank-1 retrieval (right; raw euclidean, pool 1,000) against training-set size. Dashed: generic boundary-token control. Layer 19.

- **The mapping has high predictive power**: held-out R^2 0.75, rank-1 retrieval 0.98 at 1 million contexts
- **The mapping is mostly linear**: the nonlinear (MLP) map adds only ~ 0.06 R^2 over the linear map, and only at large n (potentially this advantage would grow as you scale up the number of contexts)
- **The mapping is not present for generic boundary tokens**: held-out R^2 0.75 for a mapping from generic boundary tokens (punctuation) to the next stretch of text
- **The mapping is not just copying the context vector**: The mapping does much better than identity + bias
- We study the **linear** map here; we are most excited about what it is *useful* for.
- **The mapping is strongest at mid-late depth**: held-out R^2 peaks at **layers 19–20 of 28** (0.76)

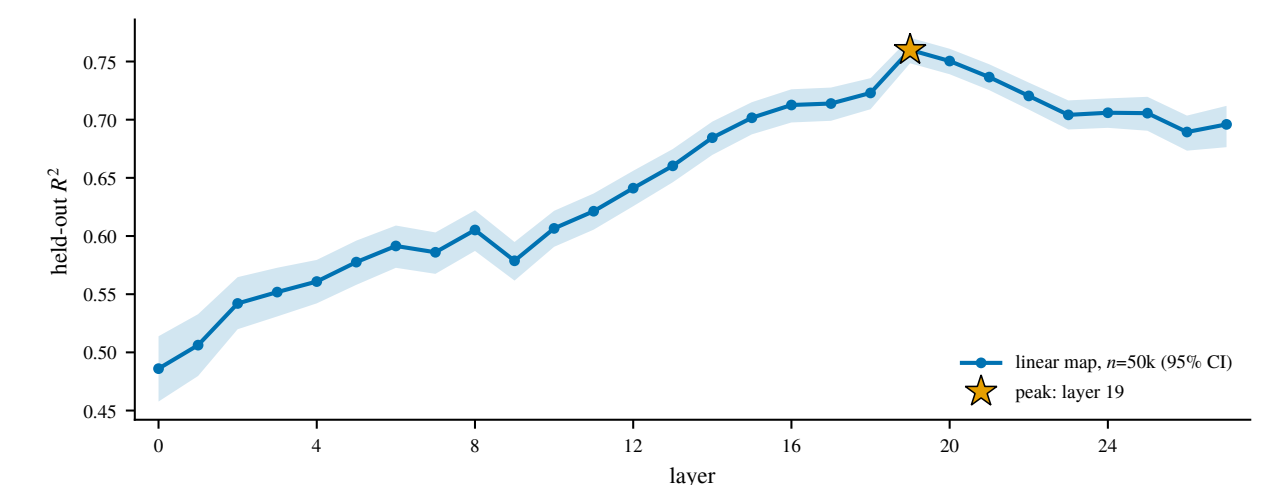


Figure 3. Held-out R^2 against the layer at which *both* v_C and v_A are taken (swept together). Band: bootstrap 95% CI over test rows. Star: argmax, layer 19 / 19 and 20 overlap. $n_{\text{train}} = 50,000$, $n_{\text{test}} = 1,000$.

2. THE MAPPING PREDICTS USEFUL DIRECTIONS OF THE ANSWER

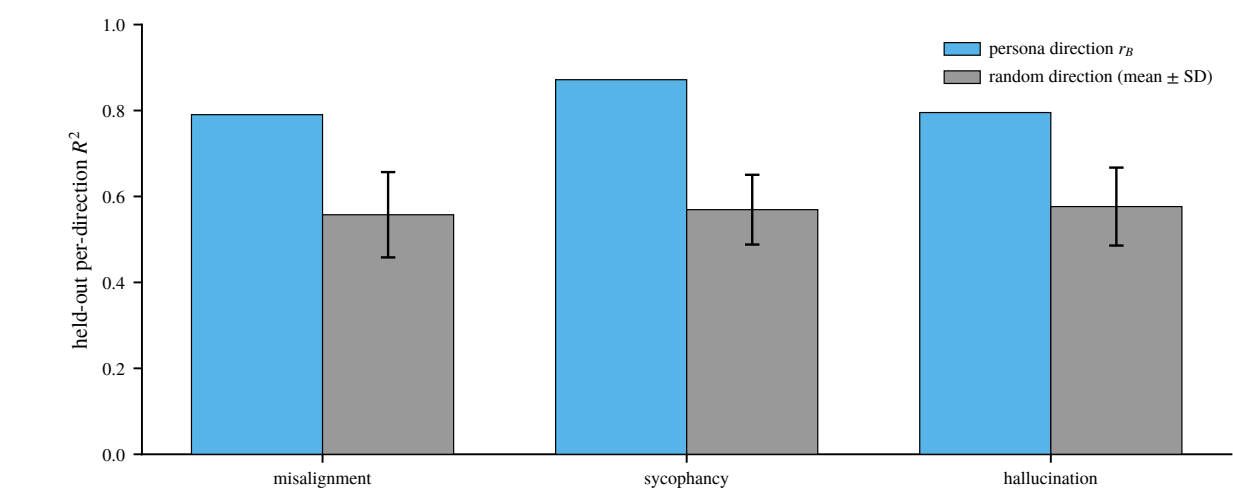


Figure 4. Held-out R^2 along each persona direction r_B against one pooled reference over all 150 random unit directions (mean $\pm$ 1 SD, pooled across the three read-out layers). Fold 0 of 5; layers 14 / 26 / 17.

- **Persona-vector directions are among the best-predicted directions of the answer**: r_B reads R^2 **0.79 / 0.87 / 0.80** (misalignment / sycophancy / hallucination) against a random-direction band of **0.56–0.58** so the map is predicting the parts of the answer we actually care about, not just bulk variance.

3. WHAT MAKES A FEATURE PREDICTABLE?

We map the context to the answer’s **SAE features**, then partial out predictors *one at a time* to see which survives.

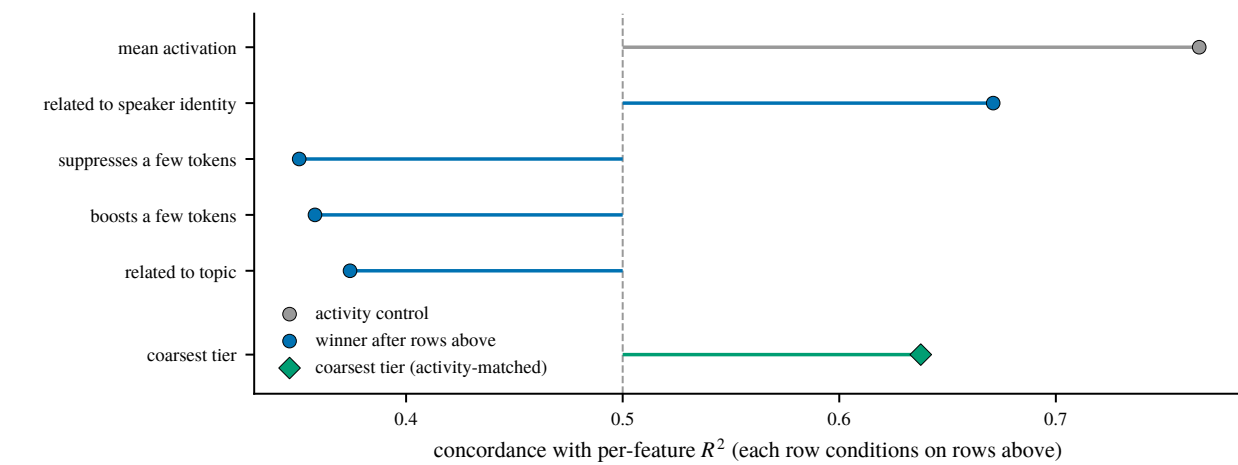


Figure 5. Forward-stepwise partialling: each row conditions on all rows above. c = concordance (AUROC scale), 0.5 = chance. Gray: activity control. Diamond: matryoshka *coarsest* tier different dictionary, direction only. No CIs; later rounds omitted.

- **Persona survives the partialling**: once both activity controls are removed, features **related to speaker identity** are the strongest surviving predictor (c 0.67), and the matryoshka *coarsest* tier points the same way (0.64).
- **Low-level and content features are predicted worse than chance**: features that **boost or suppress a few output tokens** (0.36, 0.35) and features **related to topic** (0.37) all fall below 0.5.
- **Concretely**: best-predicted are abstract nouns and nominalized concepts (R^2 0.95); worst are 2–4-character word fragments and single-letter detectors ($R^2 < 0$).

4. WHAT DOES IT FAIL TO RETRIEVE?

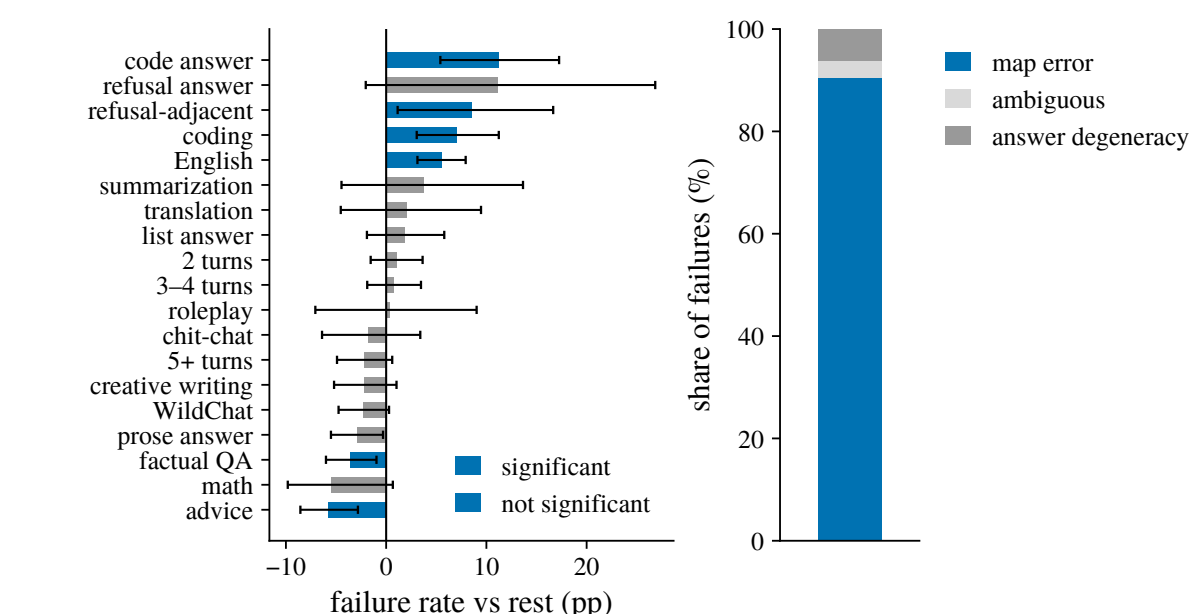


Figure 6. Failures against 5-draw-averaged answer targets, raw euclidean: $n = 1,988$, pool 9,941, 180 failures, rank-1 0.909. Blue clears BH FDR at $q = 0.05$. Right: what the misses are.

- **Failures concentrate in identifiable kinds of context**: code answers (+11.2pp), refusal-adjacent requests (+8.5), coding (+7.0) and English (+5.6) fail most; factual QA (−3.5) and advice (−5.8) fail least.

5. WHERE DOES THE MAPPING COME FROM?

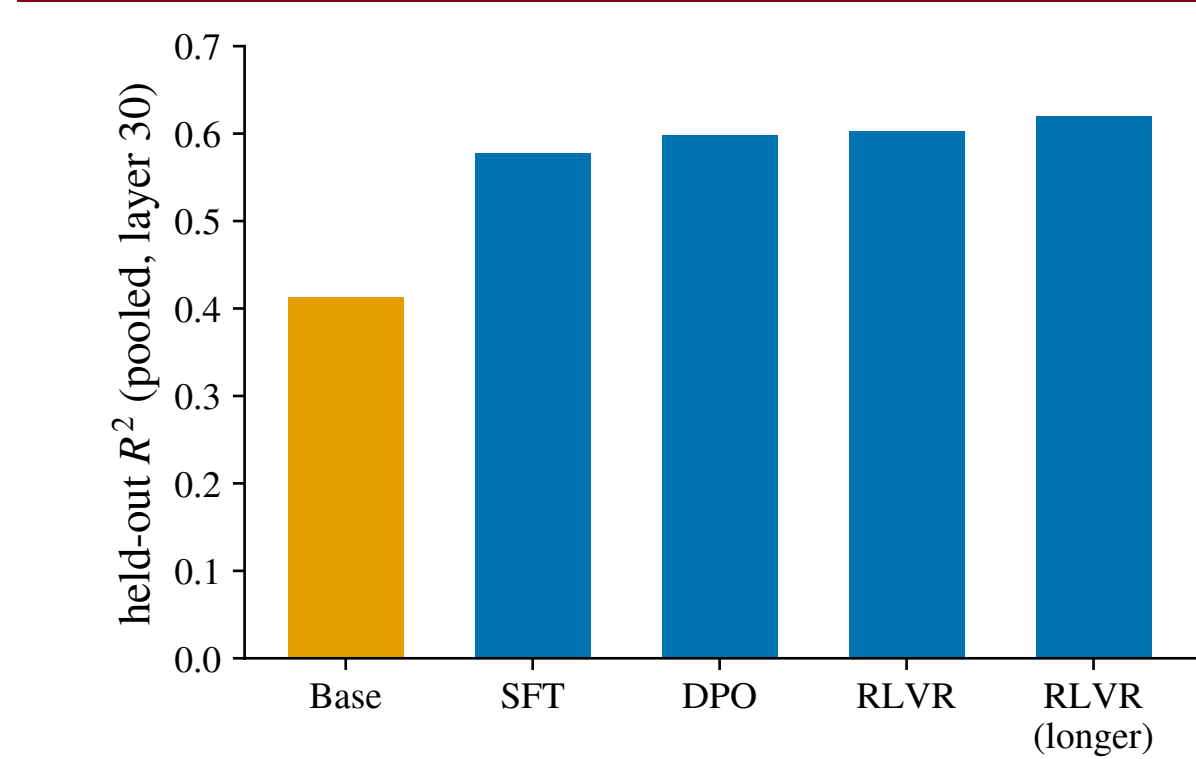


Figure 7. Pooled held-out R^2 at each stage of the Llama/Tulu ladder (layer 30), each stage fit and evaluated on its own on-policy answers.

- **Already largely present in the base model**: the base bar reads R^2 0.41 before any post-training.
- **SFT is by far the largest step** (+0.17); everything after it is essentially flat post-training *refines and reparameterizes* the mapping rather than creating it.

6. IS IT A PERSONA MAPPING?

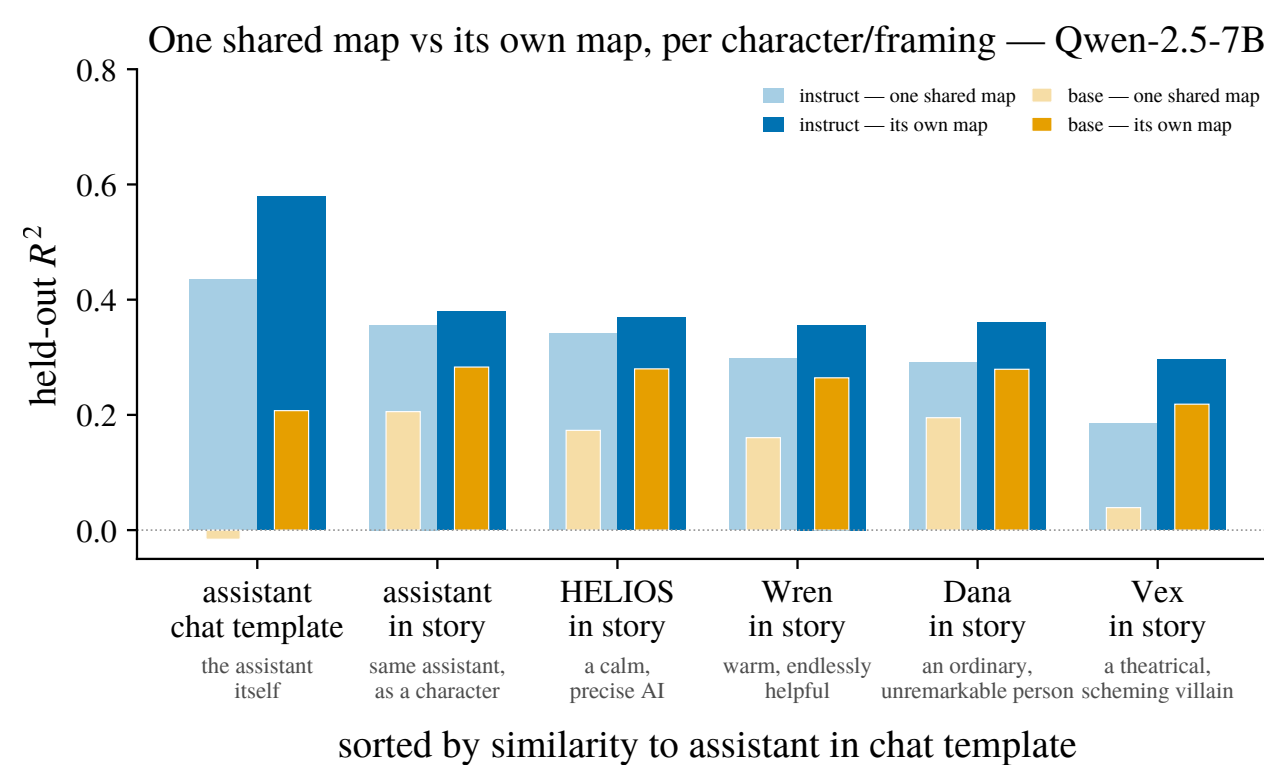


Figure 8. Held-out R^2 per character/framing: ONE map fit jointly across all 56 cells of the framing $\times$ identity $\times$ model lattice, against that cell’s own map. Instruct wide/behind, base narrow/in front. Story cells share the boundary name replied: "..."; the chat cell is the Qwen chat template. On-policy answers, layer 19; x sorted by how directly the assistant’s chat-template map transfers in.

- **One shared map covers most of every character’s own fit**: across the four characters and the assistant-in-prose it reaches **62–94%** of each one’s own-map ceiling (instruct) one operator, not six.
- **Being “the assistant” buys almost nothing once the frame changes**: the assistant written as a story character is the closest cell to the assistant’s chat-template map, but only barely ahead of the four characters (−0.27 direct transfer vs −0.30 to −0.51), and every one of them is negative the frame moves the map far more than the identity does.
- **Instruction tuning buys a stronger map inside its own template**: the chat cell reads 0.58 own-map on instruct against 0.21 on base, and it is the one cell where the shared map genuinely fails on base (−0.02).

7. APPLICATION: BEHAVIOR PREDICTION BEFORE INFERENCE

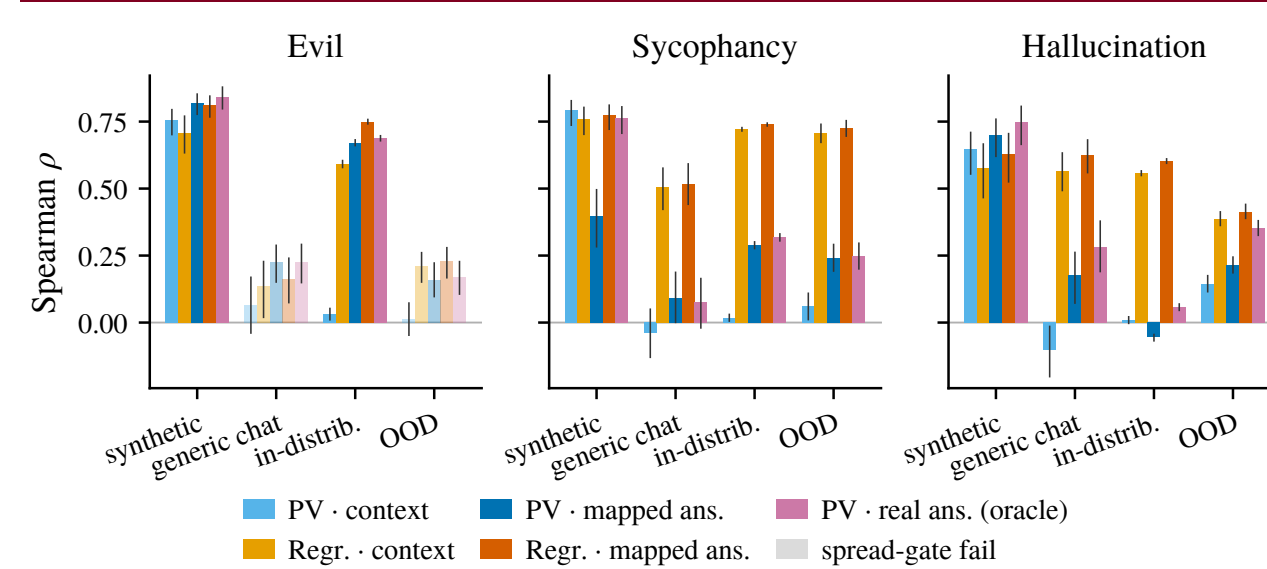


Figure 9. Spearman ρ between readout and judged behavior, per regime. Readouts: persona vector or fitted regression, on the context or the map-predicted answer; oracle = real answer. Muted bars fail the spread gate.

- **The best pre-inference readout is on the mapped answer in every trustworthy generic-chat and OOD cell**, by +0.01 to +0.07 over the best context-side readout two sycophancy margins sit inside overlapping CIs.
- **Regression on the mapped answer matches or beats regression on the context everywhere**, and both leave persona-vector-on-context near zero the gain is the *map*, not the readout family.
- **On synthetic prompts the context already gives the behavior away** the map earns its keep where the prompt does *not* telegraph the answer.

OTHER RESULTS

- **The mapping does not get reliably better with model scale**
- **The map survives a reasoning trace**: OpenThinker2-7B reads 0.78 against 0.80 for its non-thinking parent, and context→CoT→answer matches the direct map.
- **But the CoT is not a privileged span**: it adds +0.20–0.33 skill, yet a *length-matched slice of the answer’s own opening* beats it in all three thinking models (every 95% CI below zero).

IMPLICATIONS & FUTURE WORK

- Much of an LLM’s behavior is **linearly predictable from a single context vector**: a cheap pre-generation audit, and evidence for the **persona selection model**.
- Theoretical analysis + dynamical systems.
- Predict other properties: correctness.
- Predicting effect of fine-tuning.
- Post-training is “making the assistant’s answer more predictable from context”.
- Automated redteaming / jailbreaking.

REFERENCES

- [1] Chen et al. Persona vectors. 2025. [2] Betley et al. Emergent misalignment. 2025. [3] Marks et al. The persona selection model. 2026. [4] Lu et al. The assistant axis. 2026. [5] Hosseini & Fedorenko. Trajectory straightening. 2023. [6] Park et al. The linear representation hypothesis. 2024.